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► Languages & Libraries

▼ Tools & Simulators ⓘ (<http://eda-playground.readthedocs.org/en/latest/intro.html#tools-simulators>)

Icarus Verilog 12.0



Compile Options

-Wall -g2012

Run Options

Run Options

- ☐ Use **run.bash** shell script
- ☐ Open **EPWave** after run
- ☐ Show output text file
- ☐ Show output SVG file
- ☒ Download files after run

▼ Examples

using EDA Playground (/playgrounds?searchString=EDAPEX&language=&simulator=&methodologies=&curated=true)

VHDL (/playgrounds?searchString=&language=VHDL&simulator=&methodologies=&curated=true)

Verilog/SystemVerilog (/playgrounds?

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UVM (/playgrounds?searchString=&language=SystemVerilog%2FVerilog&simulator=&methodologies=UVM&curated=true)

EasierUVM (/playgrounds?

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SVAUnit (/playgrounds?searchString=&language=&simulator=&methodologies=&libraries=SVAUnit&curated=true)

SVUnit (/playgrounds?searchString=&language=&simulator=&methodologies=&libraries=SVUnit&curated=true)

VUnit (Verilog/SV) (/playgrounds?searchString=&language=&simulator=&methodologies=vunit=true&curated=true)

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Python + Verilog (/playgrounds?

searchString=&language=Python+%2B+SV%2FVerilog&simulator=&methodologies=&curated=true) Python + VHDL

(/playgrounds?searchString=&language=Python+%2B+SV%2FVHDL&simulator=&methodologies=&curated=true)

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C++/SystemC (/playgrounds?searchString=&language=C%2B%2B+only&simulator=&methodologies=&curated=true)

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testbench.sv



```
1 `timescale 1ns/1ps
2
3 module testbench;
```

```
4
5     reg A, B;
6     reg EN;
7
8     wire and_Y;
9     wire or_Y;
10    wire not_Y;
11    wire nand_Y;
12    wire nor_Y;
13    wire xor_Y;
14    wire xnor_Y;
15    wire tri_Y;
16
17    // Instantiate Modules
18    and_gate U1 (.A(A), .B(B), .Y(and_Y));
19    or_gate U2 (.A(A), .B(B), .Y(or_Y));
20    not_gate U3 (.A(A), .Y(not_Y));
21    nand_gate U4 (.A(A), .B(B), .Y(nand_Y));
22    nor_gate U5 (.A(A), .B(B), .Y(nor_Y));
23    xor_gate U6 (.A(A), .B(B), .Y(xor_Y));
24    xnor_gate U7 (.A(A), .B(B), .Y(xnor_Y));
25    tristate_buffer U8 (.A(A), .EN(EN), .Y(tri_Y));
26
27    initial begin
28
29
30        $dumpfile("dump.vcd");
31        $dumpvars(0, testbench);
32
33        $display("-----");
34        $display("A B EN | AND OR NOT NAND NOR XOR XNOR TRI");
35        $display("-----");
36
37        // Test 1
38        A=0; B=0; EN=0;
39        #10;
40        $display("%b %b %b | %b %b %b %b %b %b %b %b",
41                A,B,EN,and_Y,or_Y,not_Y,nand_Y,nor_Y,xor_Y,xnor_Y,tri_Y);
42
43        // Test 2
44        A=0; B=0; EN=1;
45        #10;
46        $display("%b %b %b | %b %b %b %b %b %b %b %b",
47                A,B,EN,and_Y,or_Y,not_Y,nand_Y,nor_Y,xor_Y,xnor_Y,tri_Y);
48
49        // Test 3
50        A=0; B=1; EN=0;
51        #10;
52        $display("%b %b %b | %b %b %b %b %b %b %b %b",
53                A,B,EN,and_Y,or_Y,not_Y,nand_Y,nor_Y,xor_Y,xnor_Y,tri_Y);
54
55        // Test 4
56        A=0; B=1; EN=1;
57        #10;
58        $display("%b %b %b | %b %b %b %b %b %b %b %b",
```

design.sv



```
1 //=====
2 // Module 1 - AND Gate
3 //=====
4 module and_gate(
5     input A,
6     input B,
7     output Y
8 );
9     assign Y = A & B;
10 endmodule
11
12 //=====
13 // Module 2 - OR Gate
14 //=====
15 module or_gate(
16     input A,
17     input B,
18     output Y
19 );
20     assign Y = A | B;
21 endmodule
22
23 //=====
24 // Module 3 - NOT Gate
25 //=====
26 module not_gate(
27     input A,
28     output Y
29 );
30     assign Y = ~A;
31 endmodule
32
33 //=====
34 // Module 4 - NAND Gate
```



Log

Share

```
[2026-07-20 06:44:27 UTC] iverilog '-wall' '-g2012' design.sv testbench.sv && unbuffer vvp a.out
warning: Some design elements have no explicit time unit and/or
: time precision. This may cause confusing timing results.
: Affected design elements are:
:   -- module and_gate declared here: design.sv:4
:   -- module nand_gate declared here: design.sv:36
:   -- module nor_gate declared here: design.sv:47
:   -- module not_gate declared here: design.sv:26
:   -- module or_gate declared here: design.sv:15
:   -- module tristate_buffer declared here: design.sv:80
:   -- module xnor_gate declared here: design.sv:69
:   -- module xor_gate declared here: design.sv:58
VCD info: dumpfile dump.vcd opened for output.
-----
A B EN | AND OR NOT NAND NOR XOR XNOR TRI
-----
0 0 0 | 0 0 1 1 1 0 1 z
0 0 1 | 0 0 1 1 1 0 1 0
0 1 0 | 0 1 1 1 0 1 0 z
```

```
0 1 1 | 0 1 1 1 0 1 0 0
1 0 0 | 0 1 0 1 0 1 0 z
1 0 1 | 0 1 0 1 0 1 0 1
1 1 0 | 1 1 0 0 0 0 1 z
1 1 1 | 1 1 0 0 0 0 1 1
```

testbench.sv:85: \$finish called at 80000 (1ps)

Creating result.zip...

adding: a.out (deflated 85%)

adding: dump.vcd (deflated 73%)

adding: work/ (stored 0%)

adding: testbench.sv (deflated 82%)

adding: design.sv (deflated 86%)

adding: ACCELLERA_OVL_VHDL_LIB/ (stored 0%)

adding: run.sh (deflated 34%)

adding: .bash_profile (deflated 21%)

adding: .bash_logout (stored 0%)

adding: .bashrc (deflated 23%)

[2026-07-20 06:44:27 UTC] waiting for download...

Done